Bounding Box-Based 3D Mapping with UGV–UAV Collaboration for Precision Agriculture

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***Abstract*—Building 3D maps in agricultural environments is challenging due to dense vegetation, irregular terrain, lack of landmarks, and unreliable GPS. This paper proposes a Bounding Box-Based 3D Mapping method using collaboration between an Unmanned Ground Vehicle (UGV) and an Unmanned Aerial Vehicle (UAV). The method simplifies crop rows and tree canopies by enclosing their point clouds in 3D bounding boxes, fused with original UAV and UGV data, producing compact maps that preserve essential structures for autonomous navigation and trajectory planning. Evaluation in a simulated Orchard scenario shows that the method could reduce map size by up to 60% while maintaining 83.6% coverage. Multi-robot collaboration proved crucial, with the UGV contributing 74% and the UAV 26% of the merged map. Overall, the proposed method demonstrates potential and deserves further investigation in more complex agricultural scenarios.**

***Index Terms*—3D mapping, bounding box representation, UGV–UAV collaboration, precision agriculture, gazebo simulator**

I. Introduction

Building 3D maps for autonomous systems in agricultural environments is a challenging task. The difficulty arises from limitations in perception and navigation in real-world agricultural settings [1]. These challenges are amplified by dense and heterogeneous vegetation, the absence of fiducial landmarks, irregular terrain, and unreliable GPS signals [2]. In this context, developing robust mapping techniques is essential to improve the performance of agricultural robotic systems.

Traditional 3D mapping approaches—typically based on dense point clouds or voxelized representations—struggle to remain reliable under these conditions [3]–[6]. Occlusions caused by tree canopies limit the visibility of ground robots, while aerial platforms often miss structural details close to the terrain. Furthermore, dense map representations tend to impose high memory and computational demands, hindering their use in real-time agricultural navigation. These limitations motivate the need for more compact, cooperative, and structure-aware mapping strategies.

To address these challenges, a 3D mapping algorithm is proposed in which collaboration between an Unmanned Ground Vehicle (UGV) and an Unmanned Aerial Vehicle (UAV) is integrated using bounding box (BB) structures. Through this collaborative strategy, the reconstruction of upper vegetation regions is improved, and compact maps are generated to facilitate trajectory planning in orchard-like environments and areas with dense canopy.

The proposed method was evaluated in a simulated Orchard scenario specifically built for this study. The environment consists of rows of simulated trees forming structured corridors that mimic agricultural plantations, providing clear navigation paths and controlled conditions for assessing the performance of the mapping approach.

In this context, the main contributions of this paper are:

- A novel Bounding Box-Based 3D mapping method with UGV–UAV collaboration for precision agriculture, designed to generate compact and informative maps of agricultural environments.
- Insights from a numerical case study in a simulated orchard environment (Gazebo), showing that the method can reduce map size while retaining essential coverage for UAV–UGV navigation and bounding box generation.

The remainder of this paper is organized as follows. Section II reviews recent studies on UGV–UAV collaboration for 3D mapping. Section III describes the proposed approach for 3D map generation based on bounding boxes and UGV–UAV cooperation. Section IV presents a case study that demonstrates the application of the proposed methodology, while Section V analyzes and discusses the obtained results. Finally, Section VI concludes the paper by summarizing the main findings and outlining directions for future research.

II. Related Work

This section reviews recent UGV–UAV collaboration approaches for 3D mapping, highlighting their objectives, key features, strengths, and weaknesses, followed by an analysis of their implications for navigation and decision-making.

A. UGV-UAV collaboration for 3D mapping

Recent work has introduced several frameworks for collaborative 3D mapping using heterogeneous UGV–UAV systems, especially in challenging environments. Several studies focus on GNSS-denied scenarios: Yue et al. [7] integrate continuous–discrete mapping with Bayesian fusion for adaptive map merging, while Wang et al. [8] combine octomap-based reconstruction with ORB-SLAM2, achieving reliable navigation indoors. Surmann et al. [9] and Zhang et al. [10] fuse UAV and UGV point clouds using planar registration or multi-layer optimization pipelines, demonstrating robust localization and mapping, albeit with sensor or environmental limitations.

Other approaches integrate visual and LiDAR data for collaborative mapping. He et al. [11] combine ego-motion estimation, pose graph optimization, and neural network-based data association, while Xie et al. [12] introduce hierarchical planning with SLAM modules for multi-robot outdoor mapping. Both achieve high accuracy but require significant computational resources.

Focusing on agricultural environments, Tagarakis et al. [6] fuse UAV orthomosaics and UGV RGB-D data to estimate tree height and canopy volume, whereas Tian et al. [13] employ semantic Octree fusion for multi-layered scene representation, facilitating autonomous planning. Liu et al. [14] enhance global pose estimation using distributed SLAM with semantic–geometric registration, and AgriColMap [15] aligns UAV and UGV vegetation data via multimodal grids and optical flow.

Finally, in dynamic or cluttered environments, Kim et al. [16] propose UAV–UGV systems for 3D data acquisition and path planning, improving efficiency and data quality despite coordination challenges. Table I summarizes key aspects of the 3D mapping methods, including sensors, alignment, merging and mapping approach, real-time capability, and main advantages and disadvantages.

B. Analysis of Heterogeneous 3D Mapping Methods

Tables I provide a representative overview of recent approaches to heterogeneous UAV–UGV collaboration for 3D mapping. A closer analysis reveals that many studies prioritize raw mapping accuracy, often at the expense of operational feasibility in autonomous navigation and real-time decision-making frameworks.

Notably, the generation of high-resolution maps enhances environmental fidelity but demands computationally intensive processing pipelines, which can hinder their deployment in real-time systems. In contrast, selective mapping strategies—such as semantic filtering and adaptive data fusion—offer a promising trade-off by emphasizing task-relevant features while significantly reducing the computational load. This balance is critical for ensuring that mapping outputs effectively support downstream autonomy functions.

Despite these advances, there remains a gap in the development of simplified mapping strategies adapted to trajectory planning and real-time decision-making. Most approaches focus on maximizing spatial detail and geometric accuracy, which often leads to excessive computational demands and reduced system responsiveness. Few studies address the challenge of maintaining a minimal yet semantically meaningful representation of the environment—one that highlights only navigable regions, obstacles, or task-relevant structures. Filling this gap requires a paradigm shift: from exhaustive reconstruction to compact and adaptive representations aligned with autonomous needs. Promising directions include online semantic segmentation, region-of-interest extraction, and adaptive resolution mapping, all aiming to enable more efficient and mission-relevant perception for robotic decision-making systems.

III. Bounding Box-Based 3D Mapping Method

The proposed method is based on the application of *Bounding Box (BB) structures* to generate a compact geometric representation of the agricultural environment by replacing crop rows with the bounding box representation of the point cloud obtained through UGV–UAV collaboration. These boxes delimit the vegetation rows and the volumes corresponding to tree canopies, providing a simplified yet information-rich spatial description of the mapped area. As illustrated in the simplified flowchart (Fig. 1), the methodology comprises sequential steps: merging point clouds, segmenting crop rows and tree volumes, organizing data into vertical layers, extracting and fusing 3D bounding boxes, and exporting a compact 3D map. Each step will be described in detail in the following subsections.

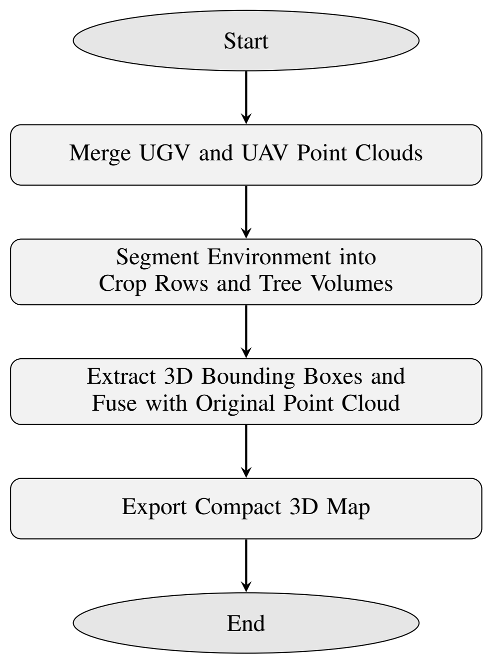


Fig. 1. Simplified flowchart of the proposed 3D mapping methodology based on bounding boxes.

A. Merge UGV and UAV Point Clouds

In this initial step, the point clouds acquired by the UGV and UAV are brought into a common reference frame and merged into a unified spatial dataset. After an optional voxel downsampling step to reduce redundancy, the alignment between the two clouds is refined using the Iterative Closest Point

TABLE I
SUMMARY OF ARTICLES ON HETEROGENEOUS COLLABORATION FOR 3D MAPPING.

Ref.	Sensors / Robots Team	Alignment, Merge and Mapping Approach	Real-time	Advantages and Disadvantages
[7]	3D LiDAR (UGV), RGB cameras (UAV)	LOAM (UGV) and ORB-SLAM (UAV); Bayesian probabilistic fusion; Hybrid SLAM (LiDAR + vision) continuous-discrete	No	Robust in GNSS-denied environments; multi-resolution fusion improves accuracy / High computational cost; sensitive to UAV visual noise
[8]	Stereo and monocular RGB cameras, IMU, proximity sensors (UGV)	Visual feature matching, 3D homography; Visual map fusion via pose graph; Visual SLAM (ORB-SLAM + inertial data)	Yes	High accuracy in indoor environments; robust to GNSS absence / Sensitive to lighting; requires stable communication
[9]	RGB cameras (UAV), 2D LiDAR and IMU (UGV)	Planar segment correspondence; Global pose optimization (pose graph); Vision + LiDAR SLAM	Yes	Integrates aerial and ground data; detailed maps / Dependent on planar surfaces; difficult in cluttered environments
[10]	3D LiDAR (both), monocular RGB camera, low-cost IMU	Multi-layer optimization with ICP and pose correction; Incremental fusion via pose graph; Visual-LiDAR SLAM with 6-DOF reconstruction	Yes	High mapping speed (up to 15 m/s UAV); robust integration / Requires accurate sensor calibration; issues with dense vegetation
[11]	RGB cameras, 3D LiDAR, IMU	Visual-LiDAR odometry with geometric constraints; Fusion via pose graph and neural networks; Visual-LiDAR SLAM with segmentation	Yes	Robust in complex urban environments / High computational cost; sensitive to reflective surfaces
[12]	3D LiDAR, RGB cameras, IMU, ultrasonic sensors	3D feature-based registration (ICP + pose graph); Cooperative fusion via distributed SLAM; Online LiDAR SLAM + multi-agent planning	Yes	High exploration efficiency; reduces mapping time / Multi-agent coordination complexity; connectivity dependency
[6]	RGB-D cameras (UGV), high-res RGB cameras (UAV)	Photogrammetry (UAV) + ICP (UGV); Fusion for volume and height correction; 3D reconstruction (photogrammetry + visual SLAM)	No	High accuracy for biomass and volume estimation; superior UGV detail / Weather dependency for UAV; high computational cost
[13]	RGB-D cameras, LiDAR, IMU	ICP + semantic matching; Semantic and probabilistic merge; Active mapping, visual + semantic SLAM	Yes	Efficient air-ground integration; detailed mapping / High computational demand; difficulties in low visibility
[14]	LiDAR, RGB cameras, IMU, GPS	Distributed ICP + graph optimization; Distributed fusion of local maps; Collaborative graph-based SLAM	Yes	Robustness, scalability, efficient cooperation / System complexity; communication requirements
[15]	RGB cameras, multispectral, LiDAR, high-accuracy GPS	Visual and geometric feature matching; Semantic and geometric fusion; Visual + geometric SLAM for precision agriculture	No	High accuracy and detail; optimized for agriculture / Requires robust hardware; limited in dense vegetation
[16]	RGB-D cameras, LiDAR, IMU	ICP + visual matching; Geometric fusion for single visualization; Offline processing for 3D reconstruction	No	High accuracy; multiple integrated perspectives / No real-time mapping; offline processing dependency

(ICP) algorithm, which minimizes geometric discrepancies and improves the consistency of overlapping regions. The resulting registered point cloud provides a coherent and accurate representation of the environment, serving as the foundation for all subsequent processing stages.

B. Segment Environment into Crop Rows and Tree Volumes

The merged point cloud is first filtered using height thresholds to isolate vegetation from low-lying ground and sensor noise. To estimate the dominant orientation of the plantation, Principal Component Analysis (PCA) is applied to the filtered points. Based on this orientation, K-Means clustering is used to separate the vegetation into individual crop rows. Ground points are then estimated through a RANSAC plane fitting procedure, and the resulting ground model is used to normalize point heights. This segmented representation isolates the crop rows and tree canopy volumes, serving as the input for the subsequent bounding box extraction stage.

C. Extract and Fuse 3D Bounding Boxes

For each layer and row, the points are filtered to remove outliers using DBSCAN and percentile-based trimming. An alpha-shape algorithm computes the convex boundaries of the points in 2D. Minimum-area rectangles are then fitted and extruded into 3D prisms, forming bounding boxes that capture the shape and spatial distribution of crop rows and trees. These bounding boxes are fused with the original point cloud to preserve important features, such as individual trees, while simplifying the overall representation.

D. Export Compact 3D Map

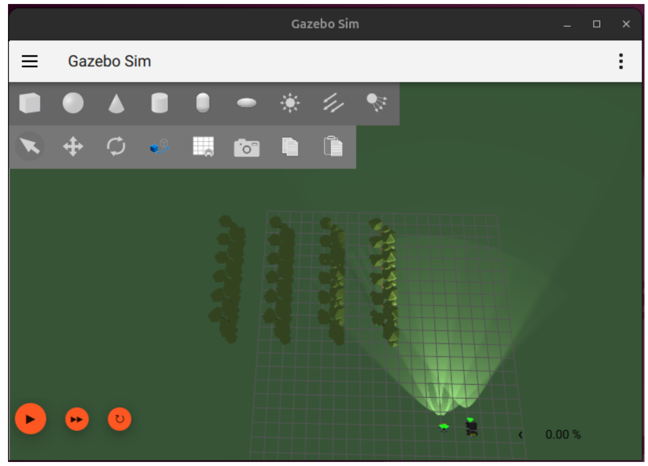
Finally, the processed point cloud, including the fused 3D bounding boxes, is exported as a PLY file. This compact 3D map provides a simplified yet information-rich representation of the agricultural environment, suitable for trajectory planning, autonomous navigation, and decision-making in cooperative robotic systems.

IV. Simulation Results

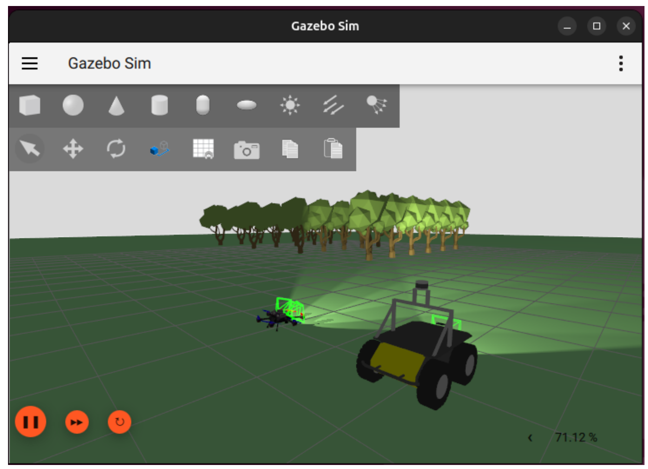
This section presents the evaluation of the proposed 3D mapping method based on BB approach. The structure has two parts: the simulation setup and the application of the proposed methodology.

A. Simulation Scenarios

The experiments were carried out in the Gazebo Harmonic simulator integrated with ROS 2 Jazzy [17]. The simulated environment corresponds to the Orchard scenario, built specifically to test the proposed methodology. It consists of rows of simulated trees forming structured corridors that replicate agricultural plantations, providing clear navigation paths and controlled conditions for evaluation. Fig. 2 depicts a top view of the scenario, including the initial positions of the quadrotor UAV and the differential-drive UGV. The UGV is a Clearpath Husky [18] equipped with a 3D LiDAR, GPS, and an RGB-D camera, enabling localization and mapping. The UAV [19] is outfitted with a 3D LiDAR, GPS, IMU, and a downward-facing RGB-D camera to support autonomous navigation and aerial coverage.



(a) Top view of the orchard scenario.



(b) UGV and UAV operating in the orchard scenario.

Fig. 2. Simulation scenario for UAV–UGV collaborative mapping: (a) Orchard top view with well-defined crop rows; (b) UGV and UAV deployed in the orchard environment.

B. 3D Mapping via Bounding Boxes

Fig. 3 illustrates the full pipeline of the proposed BB-based mapping methodology in the orchard scenario. The process begins with the acquisition of the individual point clouds from the UGV (a) and the UAV (b). These clouds are then merged into a unified 3D representation of the environment (c). Based on this combined map, bounding boxes are designed around the tree rows (d). The boxes are subsequently filtered to remove internal or redundant points, resulting in a cleaner and more structured layout (e). Finally, the UAV and UGV trajectories are incorporated into the map (f), highlighting the collaborative exploration carried out by the two robots.

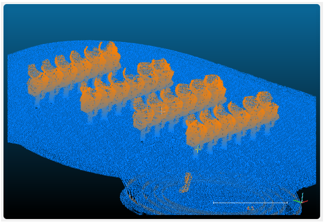
The simplified 3D map obtained through the BB-based methodology can be directly applied for autonomous navigation and planning. Fig. 4 shows an example of this application: the map provides a navigable representation of the orchard for the UGV, allowing path planning in 3D space, while the UAV can use the same map for aerial coverage and trajectory optimization. By reducing the complexity of the environment while preserving the key structural elements, the simplified map enables efficient and safe operation of both robots in a collaborative manner.

V. Discussion

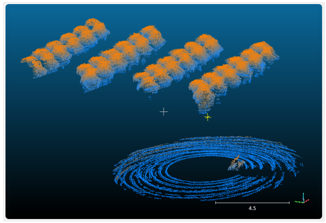
This section discusses the performance of the proposed 3D mapping methodology with respect to map compactness, spatial coverage, and the benefits of UAV–UGV collaboration. The analysis focuses on how different mapping strategies affect storage requirements and the practical usability of 3D maps in structured agricultural environments.

A. Bounding Box-Based 3D Mapping Evaluation

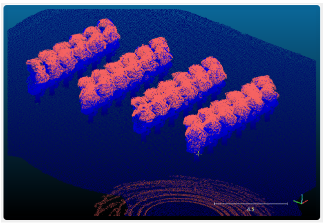
Table II presents a quantitative comparison between the original dense point cloud and the proposed BB representation in the orchard scenario. The results indicate that the BB-based approach achieves a significant reduction in map size (approximately 60%) while preserving a high XY coverage



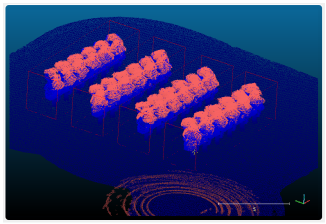
(a) UGV Point Cloud



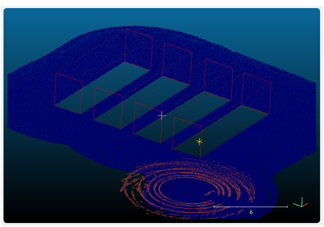
(b) UAV Point Cloud



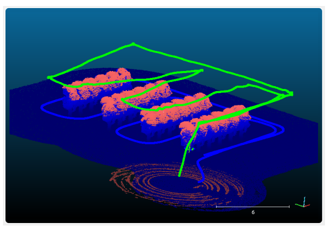
(c) UGV + UAV Point Clouds Merged



(d) Bounding Boxes Designing



(e) Filtered Bounding Boxes

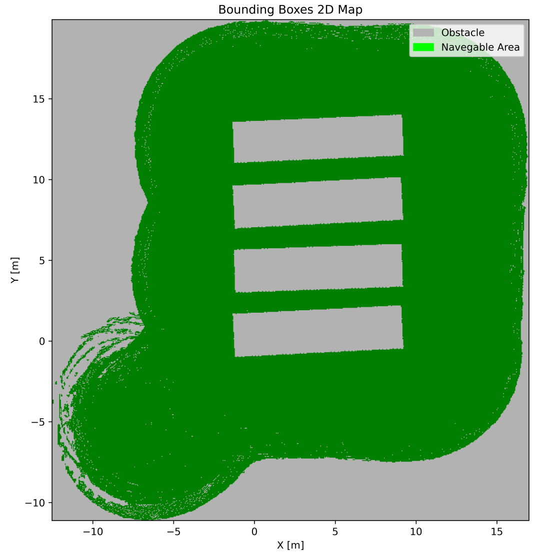


(f) Robots' Paths

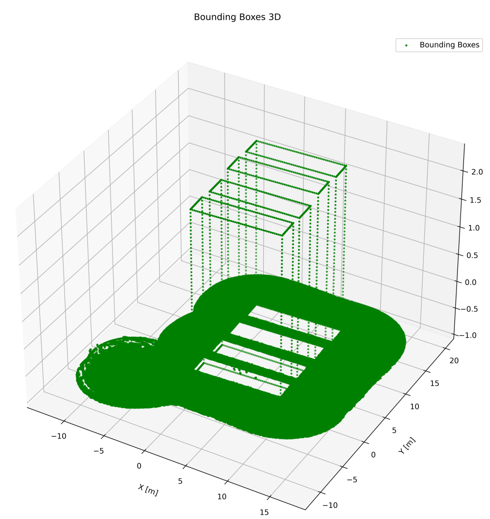
Fig. 3. Orchard scenario: (a) UGV point cloud, (b) UAV point cloud, (c) merged environment, (d) bounding boxes, (e) filtered bounding boxes, (f) bounding boxes with robots' paths.

TABLE II
ORCHARD SCENARIO: 3D MAPPING APPROACHES COMPARISON

Metric	Dense PC	Bounding Boxes
Size (MB)	14.68	5.86
Reduction (%)	-	60.09
XY Coverage (%)	100.00	83.64
Mean Z Range (m)	0.25	0.012
Max Z Range (m)	2.65	3.00



(a) UGV Simplified 3D Map for Navigation



(b) UAV Simplified 3D Map for Aerial Coverage

Fig. 4. Application of the BB-based simplified 3D map in the orchard scenario: (a) UGV navigation map, (b) UAV coverage map.

(above 83%), which is adequate for autonomous UAV–UGV navigation tasks.

Vertical metrics, such as mean and maximum Z range, further demonstrate that essential height information of trees and crop rows is preserved despite the strong reduction in point density. In particular, the bounding box abstraction captures the vertical extent of obstacles through minimum and maximum Z limits, providing a compact geometric representation while reducing sensor noise and redundancy commonly found in dense point clouds. These results suggest that the proposed BB method offers an effective trade-off between map compactness and environmental representation, making it suitable for navigation and planning in structured agricultural settings.

B. UGV-UAV Collaboration Assessment

The contribution of each robotic platform to the final 3D map was evaluated to highlight the importance of collaborative mapping. Table III shows the percentage of points contributed by the UGV and UAV, illustrating that combining both robots significantly increases coverage of the orchard environment.

Mapping with only the UGV shows a noticeable reduction in coverage, particularly in elevated areas, demonstrating the complementary role of the UAV. The contribution of both robots is especially relevant for the BB method, as collaborative mapping ensures that enough points are available to generate accurate bounding boxes for obstacles and tree rows. This analysis emphasizes that multi-robot collaboration not

TABLE III
UGV-UAV COLLABORATION IN 3D MAPPING.

Metric	**Value**
UGV contribution	74.4%
UAV contribution	25.6%
Total points in merged map	653,298
UGV-only map points	485,872

only improves spatial coverage but also enhances the reliability and usability of the BB-based 3D map for autonomous navigation.

VI. CONCLUSIONS

This paper presented a compact 3D mapping method for agricultural environments using UAV–UGV collaboration and bounding box representations. Evaluation in the Orchard scenario shows that:

- The proposed approach achieved a reduction of up to 60.1% in map size when compared to dense point cloud representations, while preserving 83.6% of the mapped environment.
- The integration of aerial and ground data increased overall spatial coverage, with the UGV contributing 74.4% and the UAV 25.6% of the merged point cloud.

The results suggest that bounding box abstraction is a promising direction for generating compact geometric representations of structured agricultural environments, particularly those characterized by repetitive layouts such as orchard rows. The use of heterogeneous aerial and ground platforms further supports the construction of more complete spatial representations by combining complementary sensing perspectives.

Future work will explore more complex agricultural scenarios and include task-level metrics for navigation and planning. It also plans to investigate the method's performance in occluded or irregular crop layouts, and to test it in real-world conditions with semantic information to better support precision agriculture.

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